

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Justification

MILU's taxonomy of 8 domains and 41 subjects covers all major board-curricular areas relevant to the deployment (Hindi Language, History, Civics, Science, Geography, Arts) and the strict MCQ format aligns precisely with the deployment's positive/negative marking requirement. However, the taxonomy was derived from competitive-exam content (1,500+ exams, predominantly UPSC/SSC/state civil-service style) rather than from school/university board syllabi. Dataset analysis confirms the framing skew: the validation sample is dominated by UPSC-style GK, current affairs, ethics scenarios, and includes highly specialised post-secondary engineering content (FORTRAN 77, AM receiver IF) that exceeds school-board level. Conversely, canonical school-board literary figures (Tulsidas, Premchand, Kabir) were absent from the sample. The 41-subject scheme appears adequate in breadth but the framing register inside subjects diverges from board syllabi.

Strengths

- Strict 4-option single-answer MCQ taxonomy is a precise structural match for the deployment's binary scoring requirement [Q22, Q33].
- Explicit inclusion of Arts & Humanities and Law & Governance as first-class domains covers civics/literature areas core to North Indian board syllabi [Q42].
- Domain-level results are reported with granular per-subject, per-language tables enabling Hindi-specific drill-down [Q98, Q99-Q132].

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment span Hindi Language and Literature (canonical authors), History (regional medieval, freedom movement), Civics (Panchayati Raj, state administration), Geography, Science, Mathematics, and General Knowledge — all in formal Hindi MCQ form. MILU's 8-domain/41-subject schema nominally covers these areas [Q42, Q23].

IO-2: The taxonomy omits a school/board-syllabus framing layer. MILU's subjects are derived from competitive-exam tags clustered into 41 subjects [Q39, Q42]; no subject category specifically represents 'Class 9-12 Hindi Sahitya' or state-board syllabus content. Canonical Hindi literary authors expected in board Literature were absent from the sample [DATASET-D3, DATASET-D33].

ID	Evidence
WEB-12	GitHub repository indicates Literature and Linguistics and Arts and Culture as subjects but no published item-level framing detail.

IO-3: Several categories include content irrelevant or over-scoped for school-board teacher use: civil-service ethics scenarios [DATASET-D17], specialised post-secondary engineering items (FORTRAN 77, AM receiver IF) [DATASET-D45,

DATASET-D46], and time-stamped current-affairs items [DATASET-D47]. These reflect competitive-exam framing acknowledged as the source [Q24, Q92].

ID	Evidence
D17	एक वरषिठ सबिलि अधकिरी के रूप में, आपको नजिी रूप से संचालति वृद्धाश्रम के वार्षकि समारोह में अतथिबिनने के लिए आमंत्रति कयि गया है — Civil officer ethics scenario — UPSC/civil services ethics framing
D47	मई 2022 में इराक के सुलेमानयिा में आयोजति तीरंदाजी एशयिा कप 2022 स्टेज 2 अभयिान में भारत ने कतिने स्वर्ण पदक जीते? — Archery Asia Cup 2022 results — current affairs, time-sensitive

IO-4: Construct underrepresentation in school-board Hindi Literature framing and construct-irrelevant variance from civil-service GK and advanced engineering content jointly compromise content validity for the deployment, though the 8-domain breadth itself remains acceptable.

Information Gaps

- No published mapping between MILU subjects and UP/MP/Rajasthan/Bihar board syllabus chapters.
- Subject-level item counts for Hindi specifically not publicly accessible (gated dataset).

Requires Expert Verification

- Whether a Hindi-medium board teacher would judge the 41-subject schema as adequately representing Class 9-12 board curricular structure.

Remediation

Gap 1: Taxonomy derived from competitive exams (UPSC/SSC/state civil services) rather than school/board syllabi; no subject category represents Class 9-12 board curricular structure.

Recommendation: Define a deployment-specific subject mapping that aligns MILU subjects to UP/MP/Rajasthan/Bihar board syllabus chapters; downweight or exclude post-secondary engineering and ethics-scenario items that exceed school-board scope.